

SafeRide

Trusted carpooling, verified by the network

Ideation Document

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1. Executive Summary

SafeRide is a real-time carpooling platform designed specifically for sub-Saharan African metropolises (Douala, Yaoundé, Lagos, Nairobi, Dakar). Its fundamental differentiation is based on a Network Trust Score engine: before each ride, an AI agent orchestrates several CAMARA APIs via

Nokia Network-as-Code to automatically assess the driver's reliability, detecting fraud, verifying location, analyzing device status and network quality.

Unlike existing solutions (Bolt, Uber), SafeRide does not only use subjective user evaluations. It relies on objective and unalterable network signals to ensure passenger safety.

2. Problem identified

2.1 Background

Major cities in sub-Saharan Africa are experiencing an explosion in mobile transport services. The African ride-hailing market is expected to reach USD 14 billion by 2030. However, there are three critical issues holding back adoption:

- **Passenger Safety** : Fake drivers using compromised accounts after a SIM swap, leading to assaults and thefts.
- **Operational fraud** : Drivers fraudulently geolocated to accept rides without being physically present.
- **Service discontinuity** : Poor connection quality cutting off GPS tracking and passenger-driver communication.

2.2 Key data

Indicator	Value
SIM Swap Incidents in Sub-Saharan Africa (2023)	+340% vs 2020
Financial losses from mobile fraud (2023)	> 1,2 Mrd USD
Trust in ride-sharing apps	42% (GSMA 2024)
Mobile Penetration in Sub-Saharan Africa	49% (up)

3. Proposed solution

3.1 Concept central

SafeRide introduces the Network Trust Score (SCR), a dynamic score (0–100) calculated in real-time before and during each race, by automatically orchestrating 5 CAMARA APIs via Nokia Network-as-Code.

3.2 Operating Flow

1. Passenger requests a ride via the SafeRide app.
2. The SafeRide AI agent queries all 5 CAMARA APIs simultaneously.

3. A confidence score is calculated by the AI for each available driver.
4. Only drivers with a score ≥ 70 are offered to the passenger.
5. Monitoring continues during the race: any anomaly triggers an alert.

3.3 Three Levels of Trust

Score	Status	System Action
70–100	Reliable	Race allowed, standard tracking
40–69	Attention	Offered ride with passenger warning
0–39	Blocked	Driver excluded, SafeRide team alert

4. Using CAMARA APIs: Nokia Network-as-Code

SafeRide orchestrates 5 CAMARA APIs simultaneously

API CAMARA	Category	Usage SafeRide
SIM Swap MANDATORY	Identity & Fraud	Detects if the driver has exchanged their SIM within 72 hours. If so, score -60 pts automatically.
Location Verification MANDATORY	Identity & Fraud	Verifies that the driver's GPS location matches their actual network location.
Device Status MANDATORY	Network Intelligence	Verifies that the driver's device is active, non-intrusive, and non-suspicious.
Number Verification MANDATORY	Identity & Fraud	Validates that the phone number is associated with the carrier declared by the driver.
Quality on Demand (QoD) CORE	Connectivity	Ensures sufficient connection for real-time GPS tracking during the race.
Congestion Insights MORE	Network Intelligence	Detects areas of network congestion to anticipate tracking breaks and recalculate the route.
Geofencing Subscriptions MORE	Network Intelligence	Triggers an alert if the driver leaves the perimeter of the intended route.

5. Technical Architecture

5.1 Technology Stack

Layer	Technologies
Frontend Mobile	React Js: Passenger Interface + Driver Interface.

Backend API	Node.js + FastAPI (Python). Orchestration des appels CAMARA.
AI Agent	Lightweight LLM (llama) for intelligent API orchestration and scoring.
Nokia NaC SDK	Official Python SDK networkascode.nokia.io for all CAMARA calls.
Database	PostgreSQL (driver profiles) + Redis (real-time race sessions).
Notifications	Firebase Cloud Messaging for passenger alerts.
Mapping	OpenStreetMap + Leaflet.js (open-source solution adapted to the African context).

5.2 Data Flow: Network Trust Score

1. Passenger requests a ride → Backend SafeRide identifies available drivers within a 3km radius.
2. For each driver, the AI Agent simultaneously triggers (parallel calls): SIM Swap Check + Location Verification + Device Status + Number Verification.
3. Result of each API→ weighted partial score. Formula: $SCR = (SIM\ Swap \times 35\%) + (Location \times 25\%) + (Device \times 20\%) + (Number \times 20\%)$.
4. If $SCR \geq 70$: driver displayed with green badge. QoD enabled to ensure connectivity.
5. During the race: continuous monitoring every 60s. Active Congestion Insights + Geofencing.

6. Innovation Layer: Agentic AI

The SafeRide AI agent goes beyond a simple API orchestrator. It implements contextual decision logic:

- **Intelligent orchestration** : The agent decides the order in which APIs are called based on context (time, geography, driver history) to optimize latency.
- **Learning fraud patterns** : The agent memorizes regional fraud signatures (e.g. frequent SIM Swap on Friday nights in Douala) and adjusts blocking thresholds.
- **Predictive Analytics** : Cross-references Congestion Insights with GPS history to predict areas at risk of disconnection and proactively propose an alternative route.
- **Passenger Communication** : Automatically generates pop-up messages in English, French, or local languages to explain blocking decisions.

7. Expected Impact & Benefits

Beneficiary	Impact
Passengers	Reduction of safety incidents related to fake drivers. Smooth ride thanks to the transparency of the trust score.
Legitimate drivers	Enhance their network reliability. Priority access to races thanks to a high score.
Mobile operators	Valuation of CAMARA APIs. Concrete demonstration of the Open Gateway on a large scale.

African Tech Ecosystem	Replicable model for any mobile service requiring strong identity verification (fintech, health, logistics).
Target 12 months	100,000 active users in 3 cities. Fraud rate < 0.5%. NPS > 70.

8. Business Model

- **Commission per ride:** 8% of the fare (vs. 15–25% for Bolt/Uber), attractive thanks to the differentiation between safety.
- **Premium driver subscription:** 5,000 FCFA/month for priority access to races with high score guarantee.
- **B2B license :** SafeRide Trust Score API sold to other apps (logistics, delivery, health) wanting to verify their mobile providers.
- **Operator partnership :** Remuneration by the mobile operator for each API call generated via NaC (Open Gateway model).

9. Alignment GSMA Open Gateway

SafeRide is directly in line with the GSMA Open Gateway vision by demonstrating how developers, without telecom expertise, can build high value-added applications using standardized CAMARA APIs.

- **Pillar 1 Advanced connectivity :** QoD ensures continuity of service over 4G/5G networks.
- **Pillar 2 Digital Identity:** SIM Swap + Number Verification create a native network KYC layer.
- **Pillar 3 Open Innovation:** The Nokia NaC SDK enables integration in a matter of hours, without telecom expertise.
- **Pillar 4 Local Impact:** Solution designed for the sub-Saharan context (FCFA award, local languages, mixed 2G/3G/4G networks).

10. Roadmap: Hackathon → Market

Phase	Period	Objective
Phase 0	April 24 – May 10, 2026	Hackathon prototype: functional mobile app + integration of 5 CAMARA APIs via NaC simulators
Phase 1	June – August 2026	Douala pilot: 500 drivers, MTN Cameroon partnership, real network tests
Phase 2	Sept – Dec 2026	Yaoundé + Lagos extension. 10,000 users. Launch of Premium Subscription

Phase 3	2027	Expansion régionale (Dakar, Nairobi, Abidjan). Licence API SafeRide Trust Score B2B
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Conclusion

SafeRide represents a unique opportunity to simultaneously solve two critical problems in sub-Saharan Africa, urban transport security and mobile fraud, by harnessing the full potential of CAMARA's standardized APIs through Nokia Network-as-Code.

Our solution concretely demonstrates that the Open Gateway is not an abstract concept, but an operational technology capable of transforming African daily services. SafeRide is extensible, open, and designed from the ground up to create a measurable and lasting impact.

SafeRide *Every race, checked. Every journey, safe.*